Semaev first-fall archive replay and trace-witness boundary

2026-10-22 — scoped validation of `crypto-autoresearcher` main `4db6ed75`

The original eight-cell toy summary was reproduced byte for byte (SHA-256 `64f6e34ceb16bd81…54 333c99e7`). It records 70 Semaev draws with first fall at degree two and ten at degree three. The 200 support-matched null-control draws were also reproduced. No new timing, attack-cost claim, research-state promotion or performance graph change resulted.

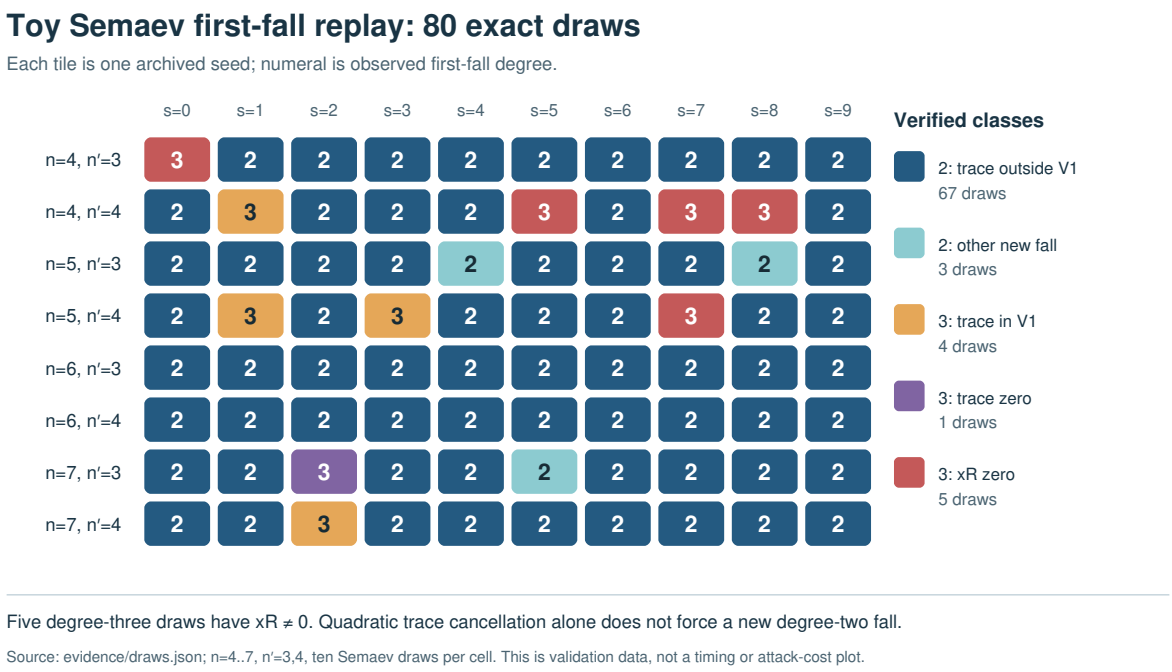


Figure: every tile is one archived Semaev seed; the numeral is the observed first-fall degree. Five degree-three draws have nonzero target abscissa, so the Lean theorem's $x_R \neq 0$ premise alone cannot establish a uniform degree-two fall. Source: `evidence/draws.json`; editable renderer: `render.py`. This is validation data, not a performance plot.

Independent checks and exact finite example

The replay checker compared 21,760 finite-field products against an independent polynomial reduction, evaluated the binary S_3 formula on all 12,800 Boolean assignments in the 80 Semaev draws, and independently recomputed the row-space rank and degree-projection rank through degree three. It also checked the support profile and ranks through degree four in all 200 null draws. All eight Semaev and null distributions match the archived table.

For $n = 4$, $n' = 4$, seed $s = 1$, the target is $x_R = 1$ and the factor-base basis is $[9, 3, 2, 12]$ over the field with modulus $X^4 + X + 1$. The trace multiplier is $\mu = x_R^{-2} = 1$. Trace cancels the quadratic part, but its nonzero linear residual $u_0+u_3+w_0+w_3$ is already the fourth input equation. In the archived Boolean filtration, $\dim V_1 = 1$ and $\dim(V_2 \cap R_{\leq 1}) = 1$, so new_falls$(2) = 0$. At degree three, $\dim V_2 = 11$ and $\dim(V_3 \cap R_{\leq 2}) = 18$, giving new_falls$(3) = 7$. The first fall is therefore three in this explicit nonzero-target example.

Among all 75 nonzero-target draws, the trace residual is a new class modulo V_1 in 67 degree-two draws. Three other degree-two draws have an old trace class but a different new relation. Four degree-three draws have a nonzero trace residual already in V_1; one has a zero trace relation. Five additional degree-three draws have $x_R = 0$ and are outside the theorem premise.

Proof boundary

The existing `DegreeTwoFall.lean` proves quadratic trace cancellation, not the nonzero quotient class required for first fall. This finite example contradicts a universal degree-two first-fall conclusion for the *archived Boolean model*; it does not refute the narrower Lean identity or establish a solving-degree or ECDLP-cost result. A faithful Lean formalization of the filtration and a checked finite rank counterexample remain open. The frozen historical target and measurement were not edited. The original 70/10 chart remains numerically current; this report adds per-draw provenance and trace classification. Full commands, source hashes, data and limits are in `report.md` and `evidence/`.

The five nonzero-target exceptions

n	n'	seed index s	x_R	Trace residual modulo V_1
4	4	1	1	nonzero polynomial, already in V_1
5	4	1	19	nonzero polynomial, already in V_1
5	4	3	19	nonzero polynomial, already in V_1
7	3	2	125	zero polynomial
7	4	2	113	nonzero polynomial, already in V_1

The five remaining degree-three draws have $x_R = 0$. Every row above has new_falls(2) = 0 under the archived definition, independently recomputed from the Boolean row spaces. Exact bases, equations, residual monomials and degree-three ranks are in `evidence/draws.json`.

Reproduction and remaining work

The original summary script was run from a scratch directory and its output compared byte for byte with the committed archive. The source SHA-256 values for the core, sampler, frozen JSON and independent checker are embedded in `evidence/draws.json`. The Python 3.12 checker is `replay_ffd_semaev_20261022.py`; its full command and output path are in `report.md`. Its field arithmetic and rank routines are independent of the archived implementations, while the sampler and equation builder are reused and checked against the direct truth table. The next formal obligation is to define the archived Boolean filtration in Lean and prove a finite rank counterexample or a precisely qualified positive theorem.